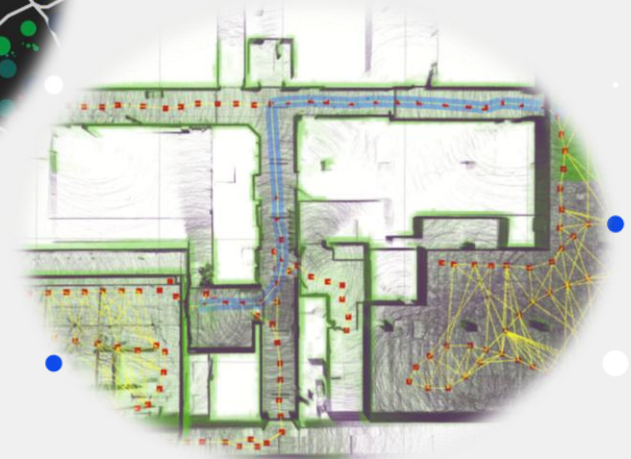
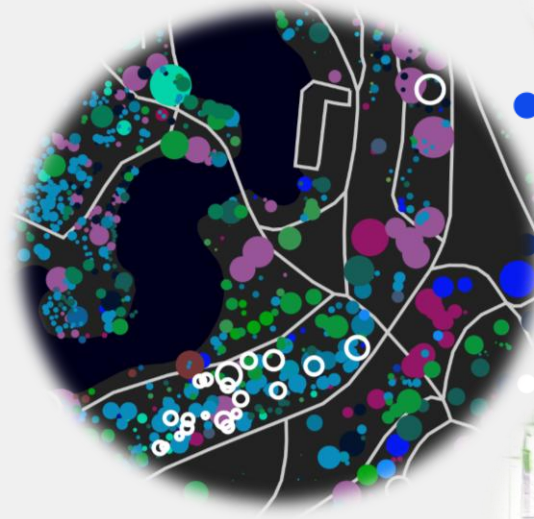
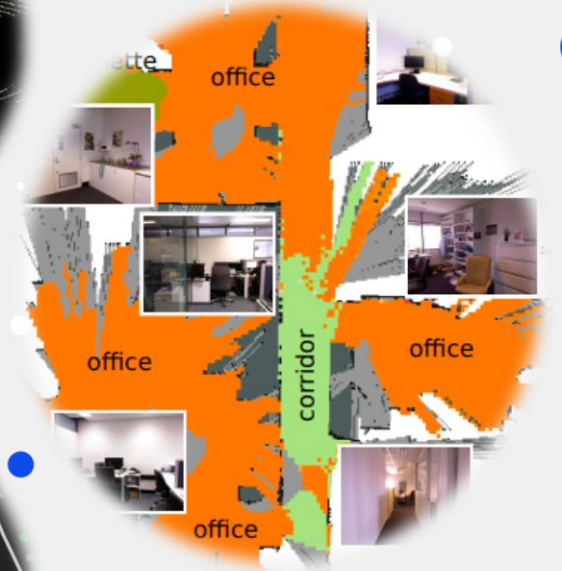
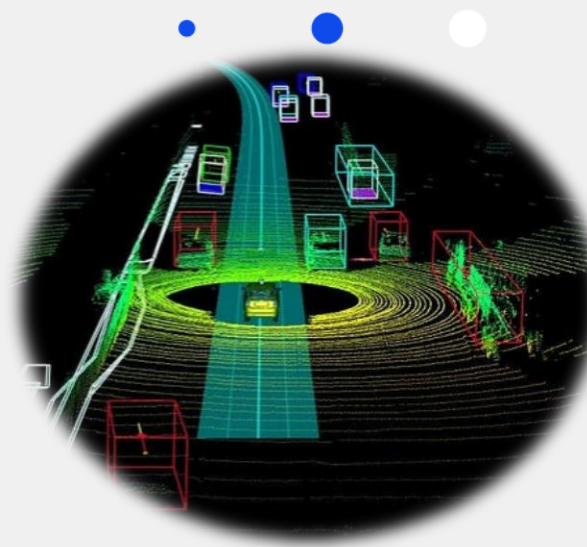




Perception and Mapping

Graeme Best
41068 Robotics Studio 1

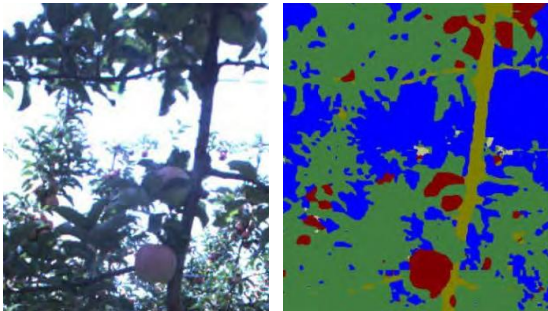
Faculty of Engineering and Information Technology
University of Technology Sydney



Robot Perception Tasks



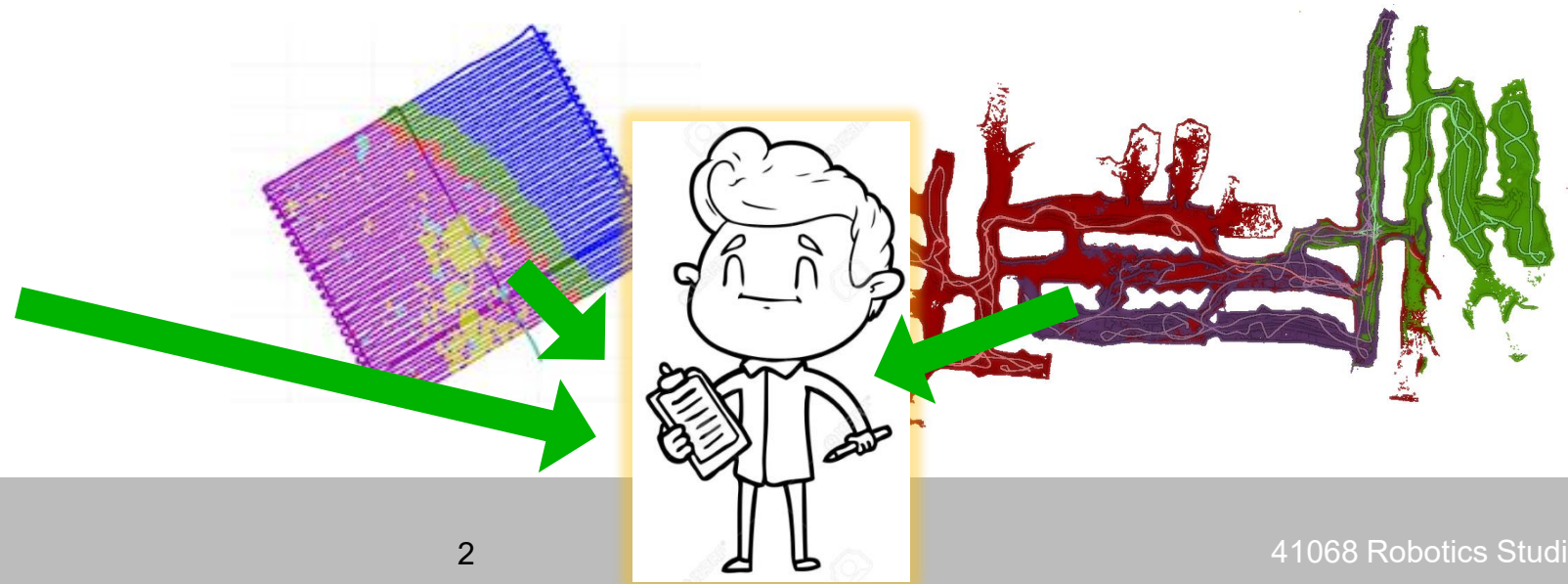
Fruit tree modelling



Ocean monitoring



Subterranean mapping

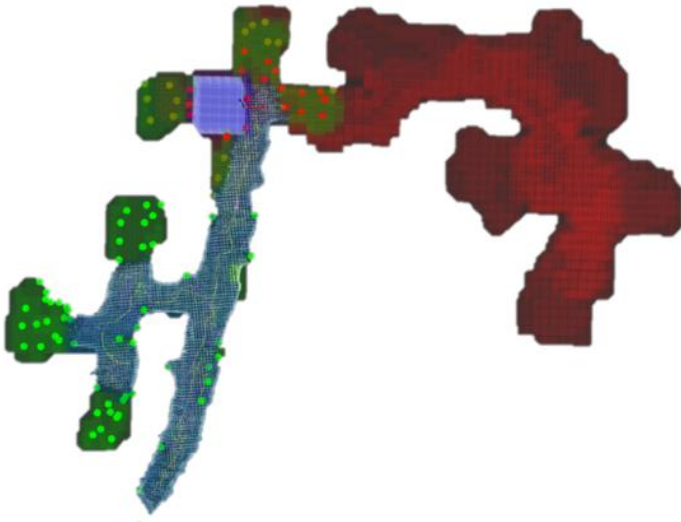


Maps for Subterranean Exploration (earlier seminar)



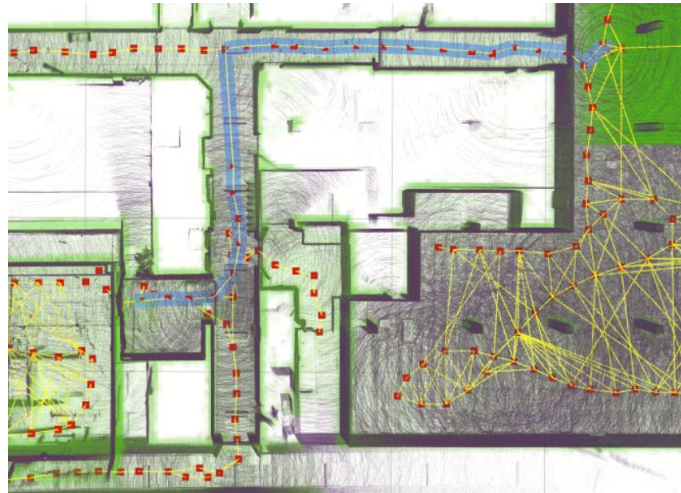
Exploration maps

- OpenVDB structure
- LiDAR- and camera-observed labels
- Shared across team



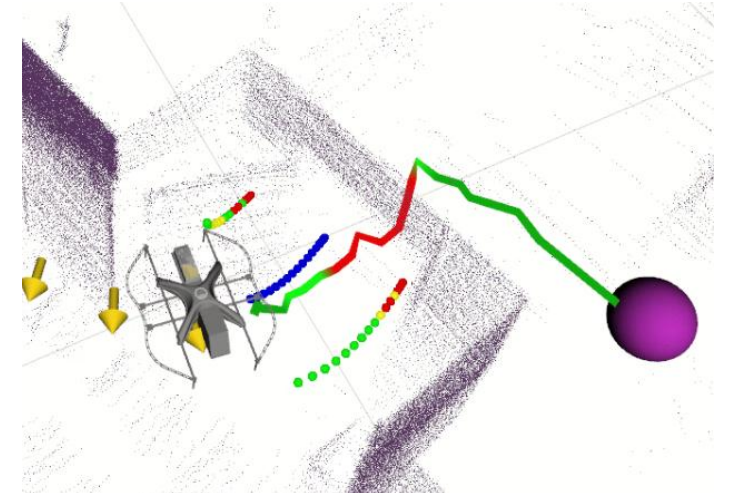
Navigation roadmap

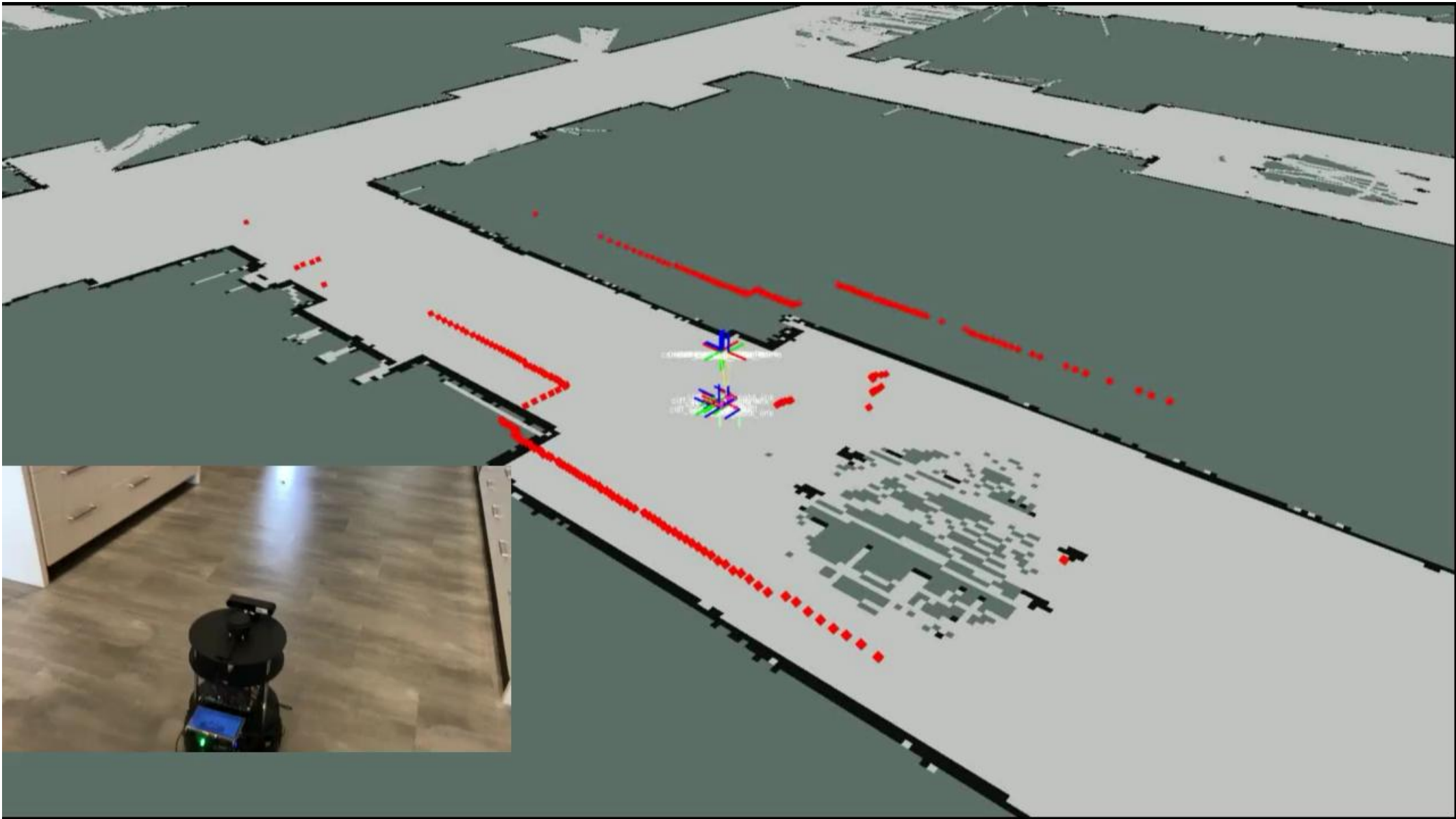
- Connect visited locations
- Repairs for changes
- Shared across team



Local planner map

- Distance transform
- Dust filtering
- Computed on GPU

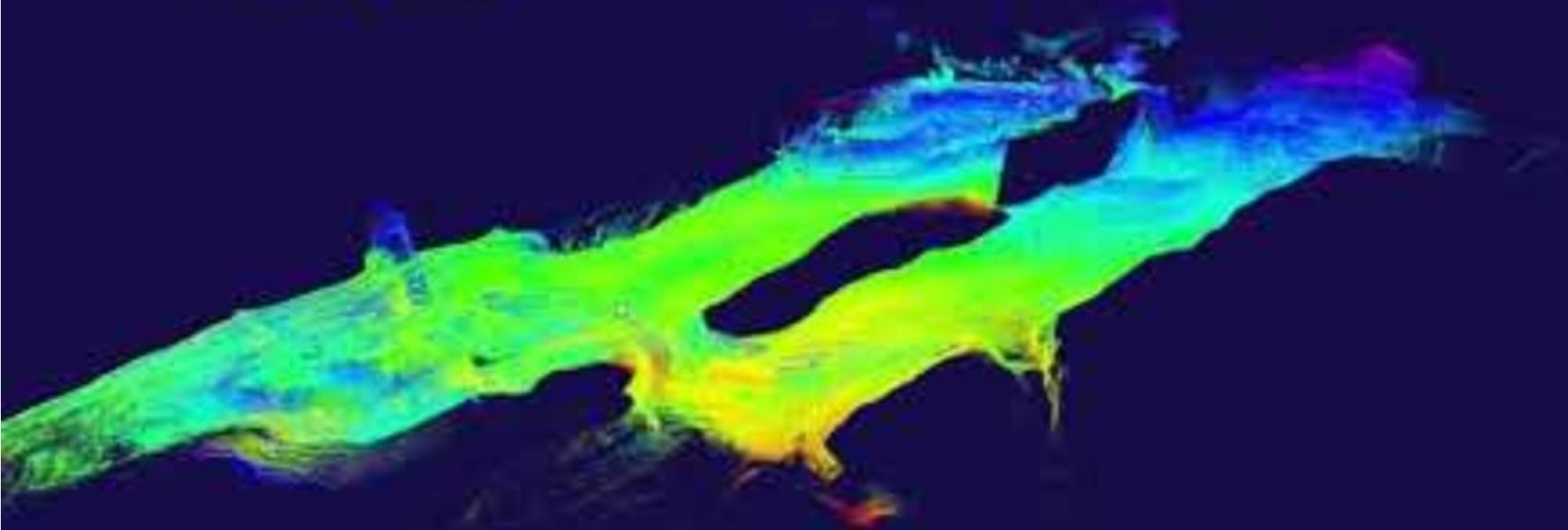






Laurel Caverns

Mapping



Questions for your 41068 Project



- Ideas for different types of maps
- Your project must have some perception/mapping
- Your project must have a user interface

- What types of maps will your robot produce?
 - For primary task
 - For autonomy
- What information will these maps contain?
- How will you show these maps in your User Interface?



Outline



- Sensors for mapping
- Map representations – many examples
- Occupancy grids
- Reflection

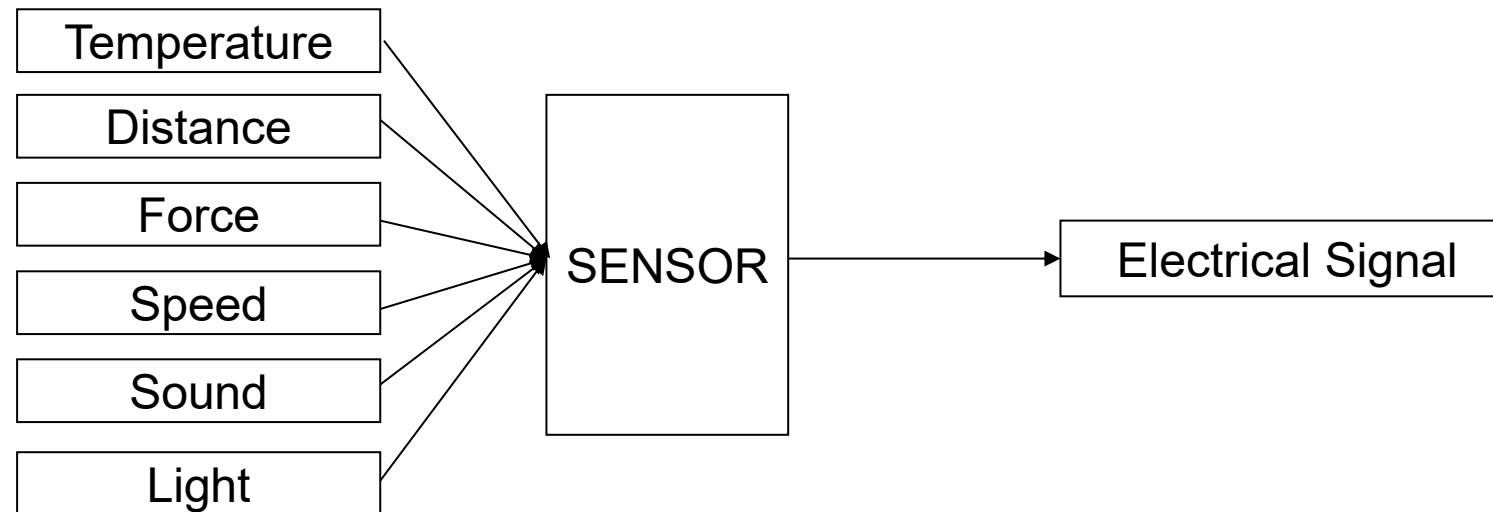


Sensors for Mapping

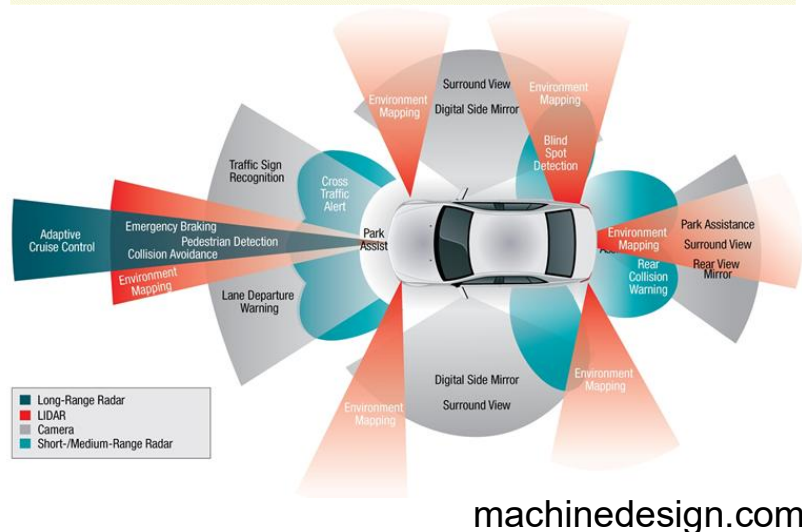
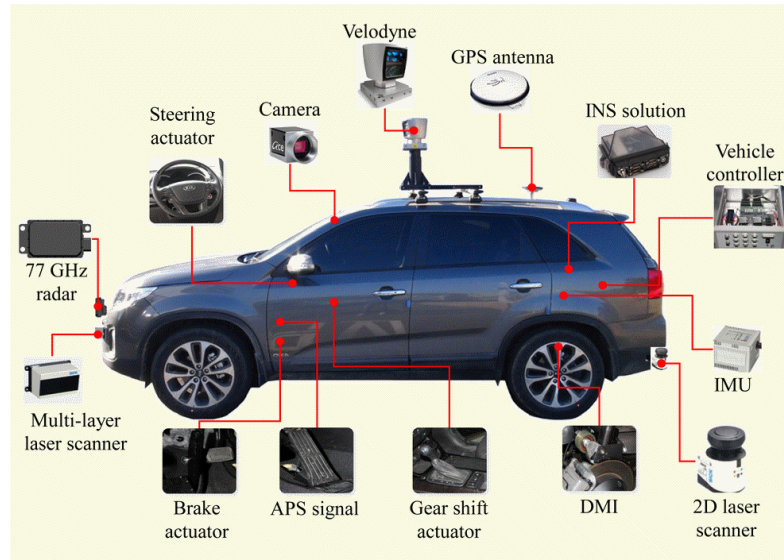
Sensors



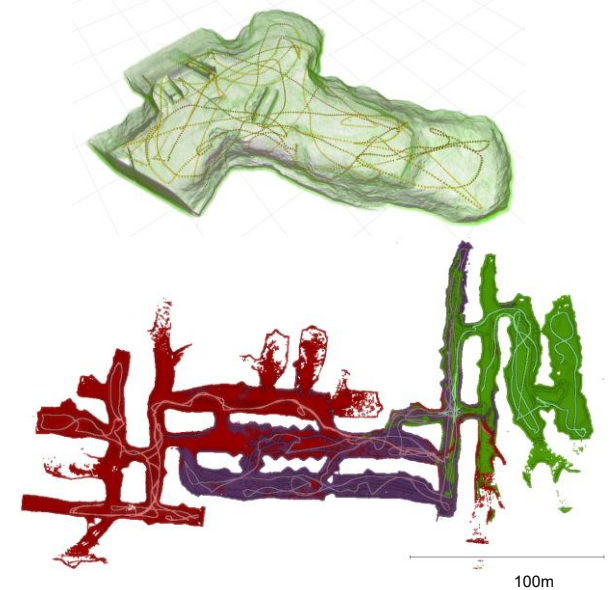
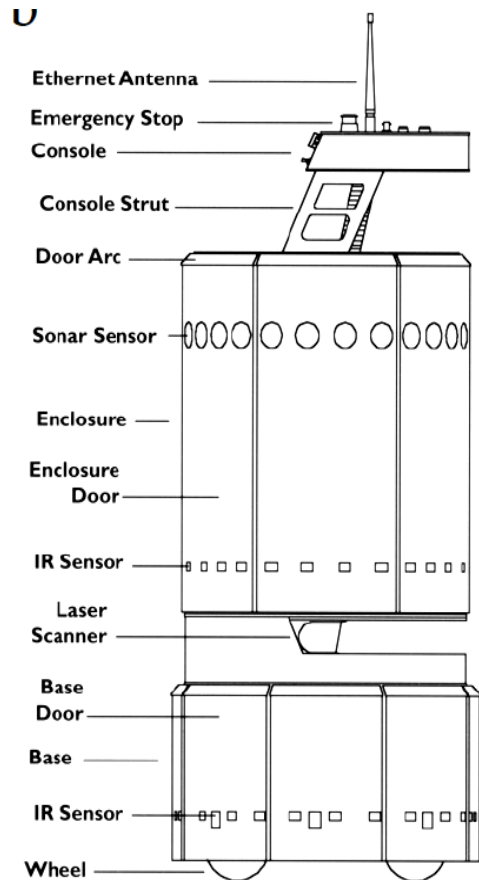
- A device for measuring some physical quantity
- The sensor usually converts from the measurement space to an electrical signal



Sensors on Autonomous Cars



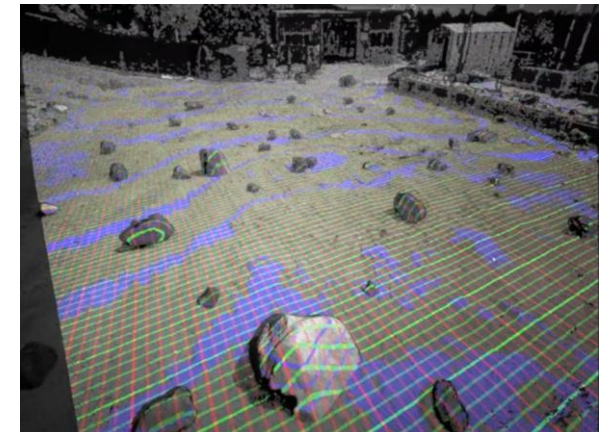
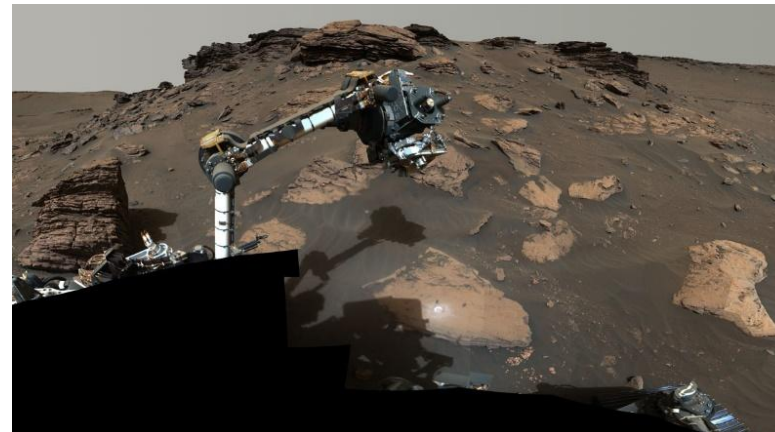
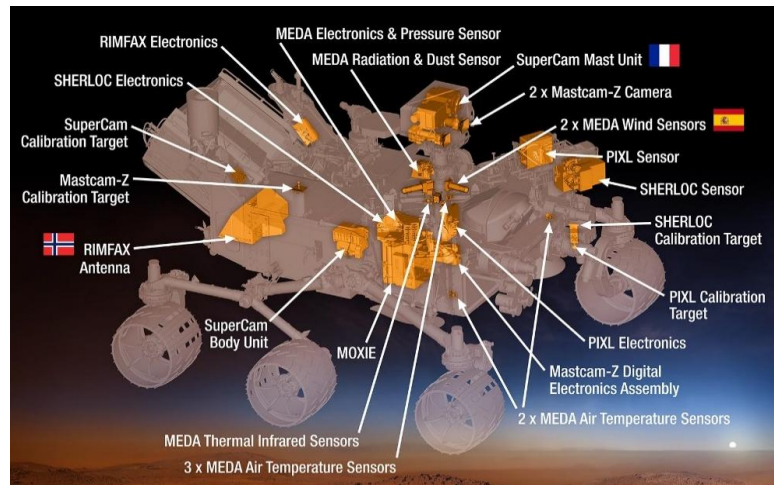
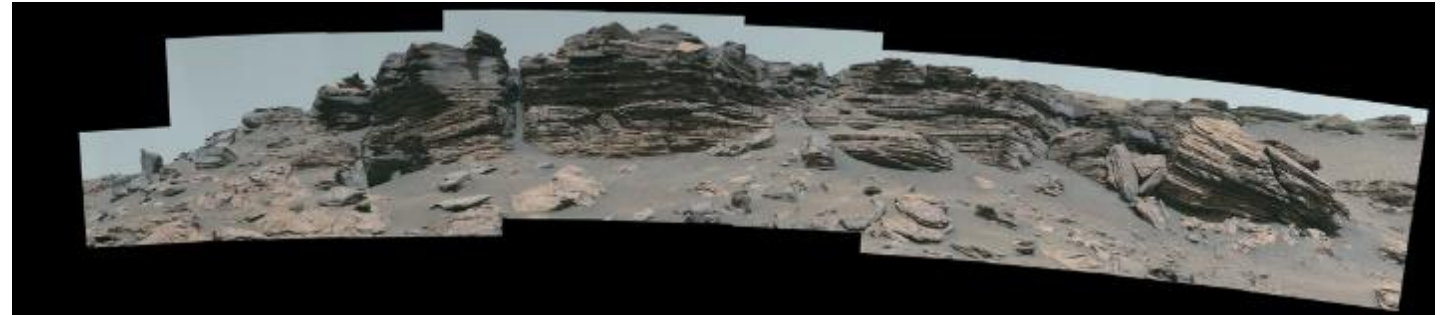
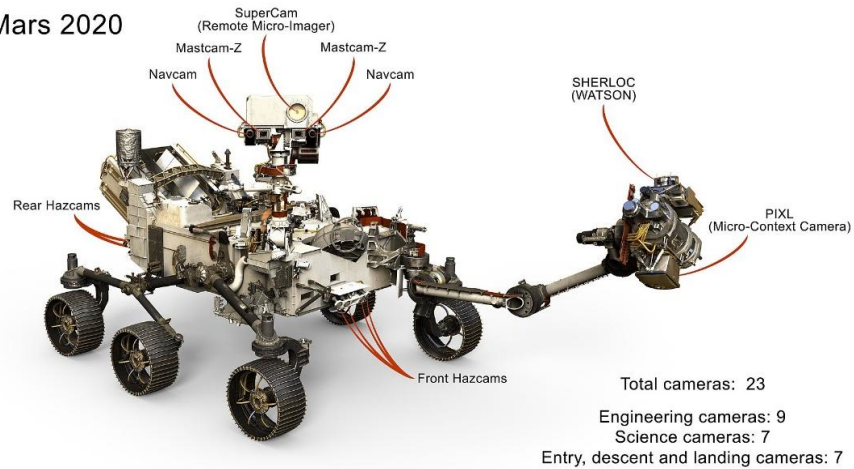
Sensors on Mobile Robots



Sensors on Mars Rover Perseverance



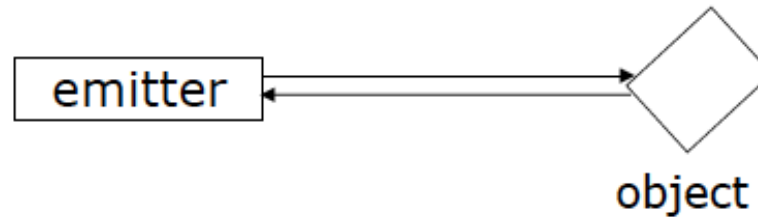
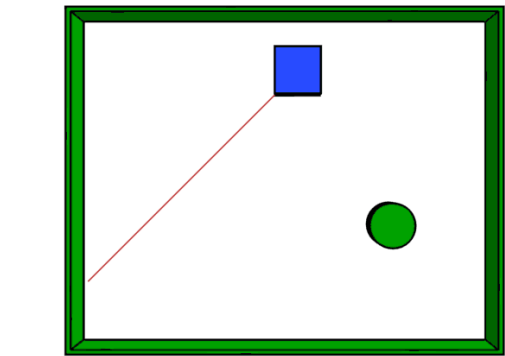
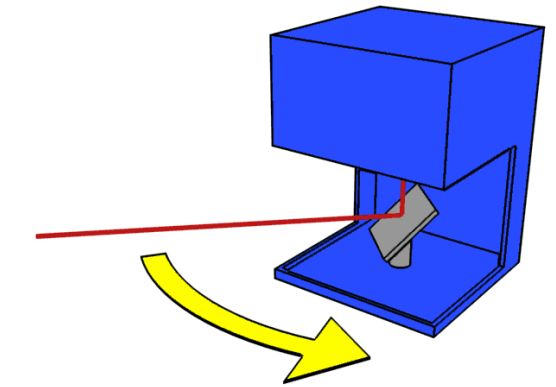
Mars 2020



See <https://science.nasa.gov/mission/mars-2020-perseverance/science-instruments/>
<https://science.nasa.gov/mission/mars-2020-perseverance/rover-components/>

LIDAR: **L**ight **D**etection **A**nd **R**anging

- Measure the time of flight of laser light pulses
- The pulsed laser beam is deflected by an internal rotating mirror
- The measurement data is available in real time via an interface





Map Representations

What is Mapping?



- **Robot mapping** is the process by which a robot builds a representation of its environment using data from its sensors (and prior information).
- This representation – often called a **map** – captures spatial information about the surroundings.
- Useful for:
 - **Enabling autonomy**: Understand location of obstacles, landmarks, traversability, navigable areas.
 - **Primary tasks**: Provide information to humans and other robots that is specific to the task. May be displayed in a **user interface**.

Why is Mapping Challenging?



- Perception challenge: Relying on sensors to understand the world
- Noisy sensors
- Local coordinates to global coordinates (transforms)
- Motion involved
- May change over time
- Many different representations, each with advantages/disadvantages

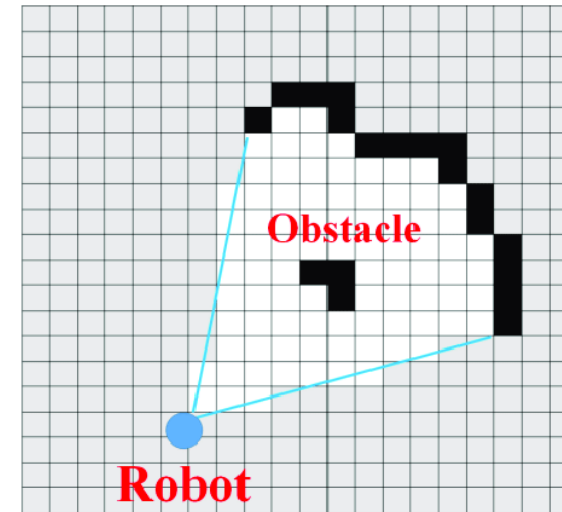
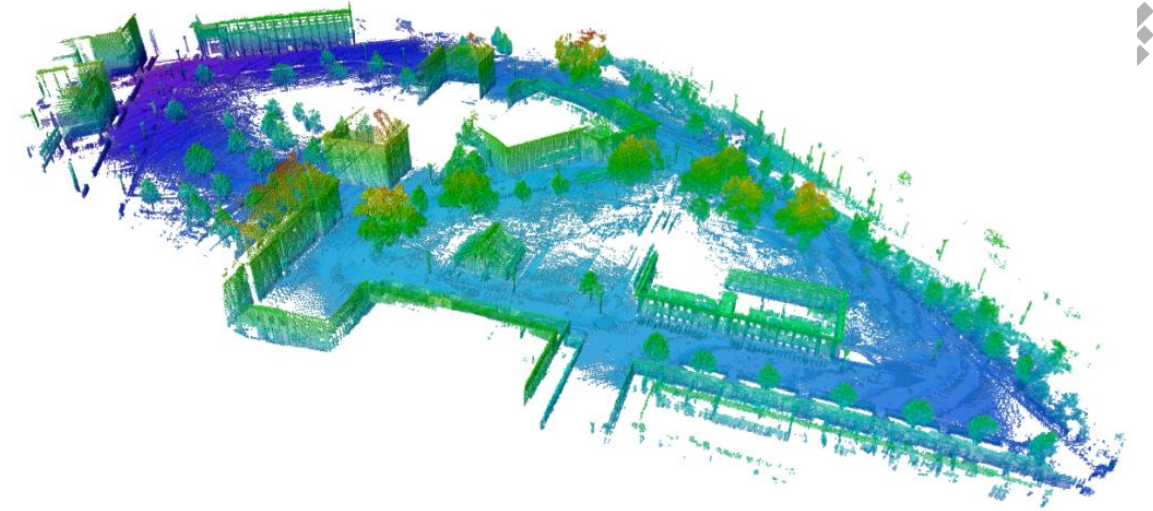
Map Categories



- Typical maps are loosely categorised as:
 - **Metric:** Accurately represents distance
 - **Topological:** Accurately represents connectivity
 - **Semantic:** Additionally associates “descriptive meaning” with locations
 - Some maps contain all of the above

Occupancy Grids

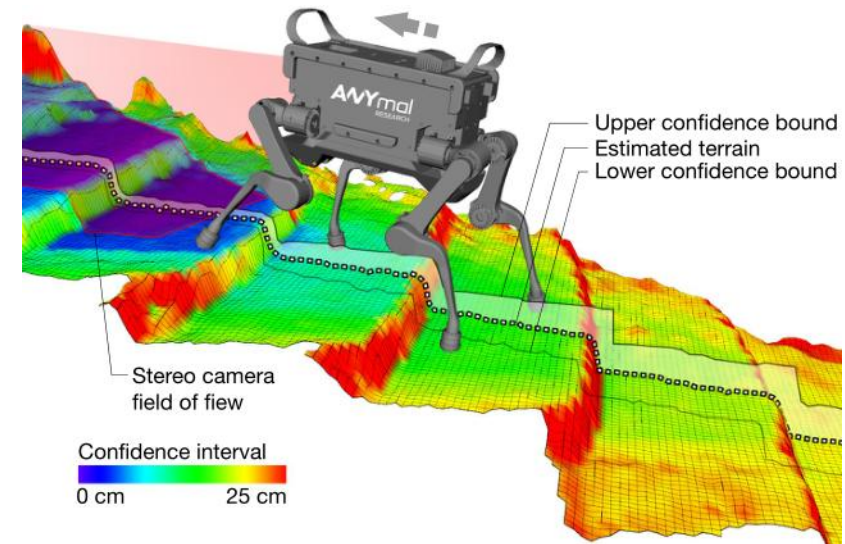
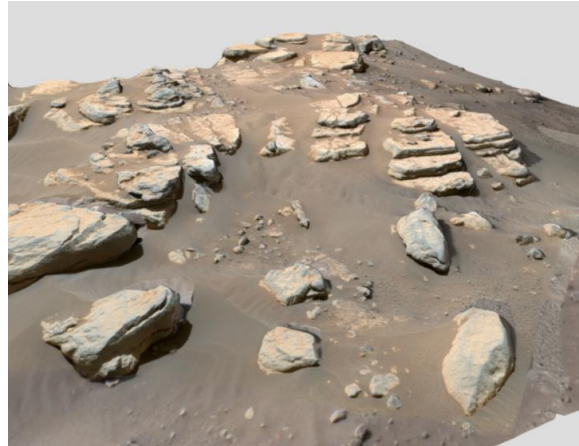
- Good for navigation
- Discretise the world into cells (2D) or voxels (3D)
- Grid structure is rigid
- Each cell is assumed to be occupied or free space
- Requires substantial memory resources
- Created from LIDAR data using ray casting
- Standard ROS packages
- Will discuss further in next section



Terrain Map



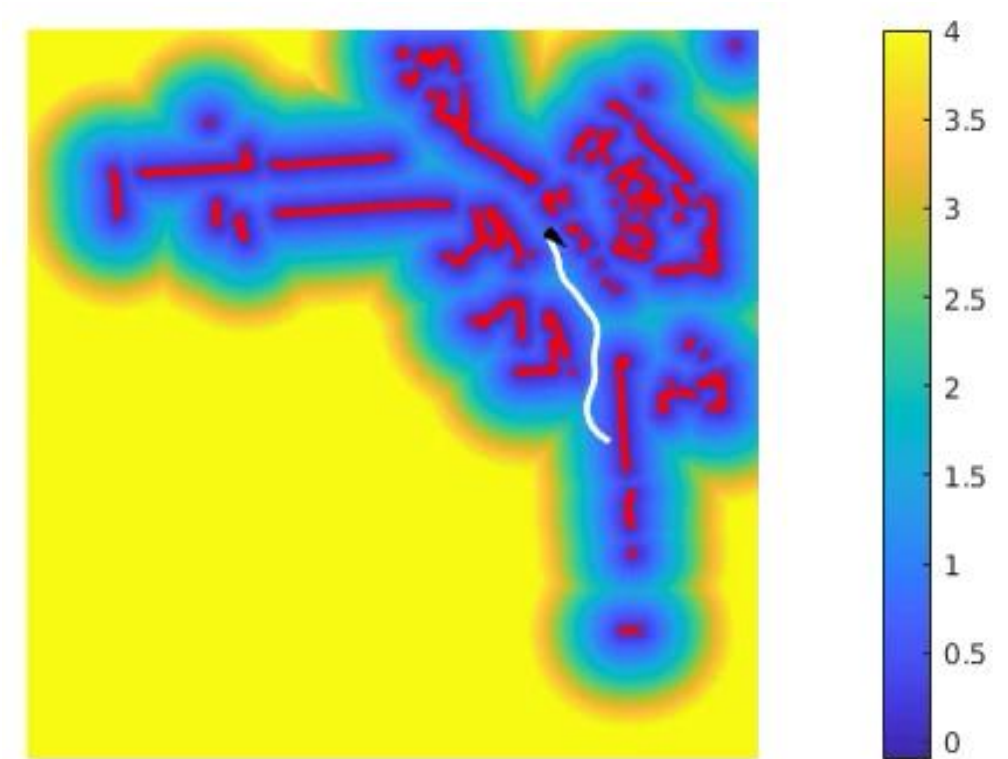
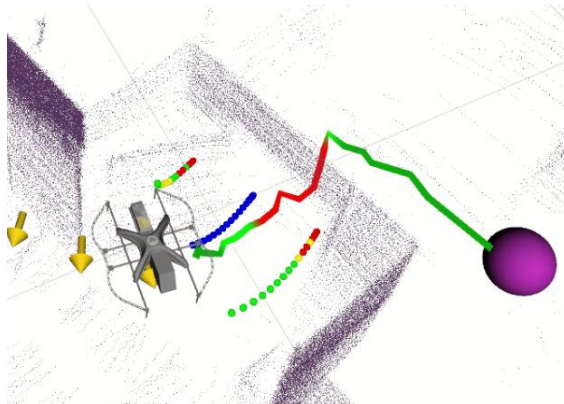
- Good for navigating complex terrain
- Similar to occupancy grid, but describes properties of the terrain, such as elevation or surface type
- Good for modelling complex 3D terrains
- Elevation often represented as a contour map



Distance Function Maps

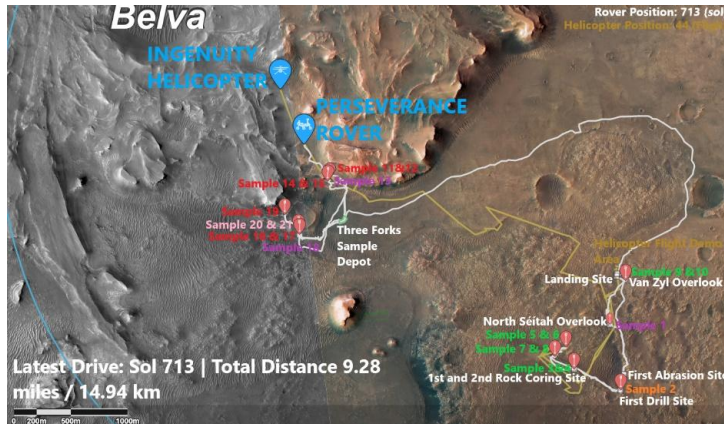


- Defines the **distance** to occupied space
- Useful for safe navigation
- Useful for fast ray tracing
- Dense representation, grid (2D) or voxels (3D)
- Expensive to compute



Feature Maps

- Good for localisation
- Landmark Based
- Kalman filter or optimisation systems (SLAM)
- Compact representation
- Multiple feature observations to improve position estimates



Landmarks on Mars



Map of trees in a park:
jillhubley.com/project/prospectparktrees/

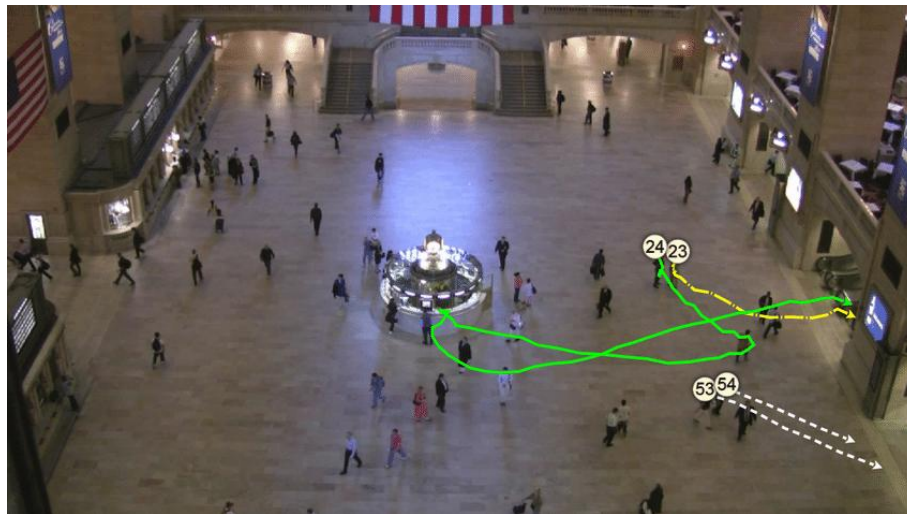


Victoria Park dataset

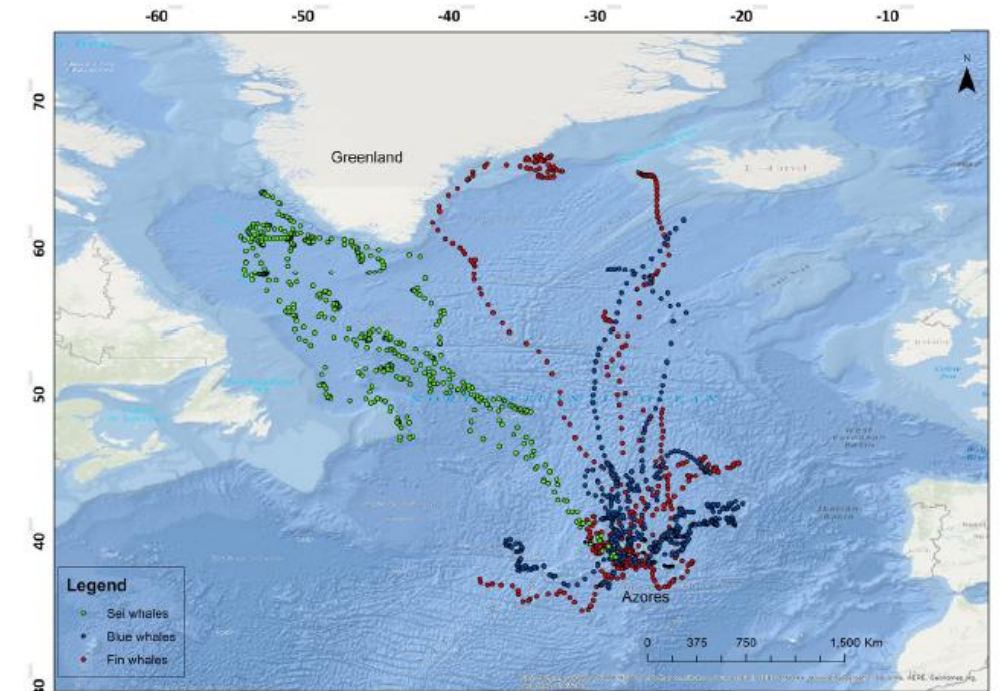
Trajectory Tracking



- Movements of objects of interest
- Similar to feature map, but the landmarks are moving over time



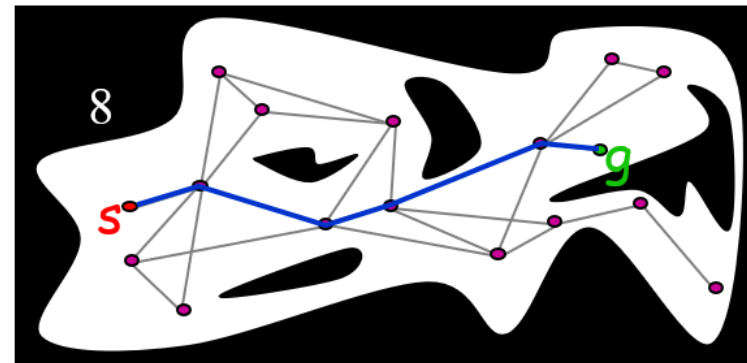
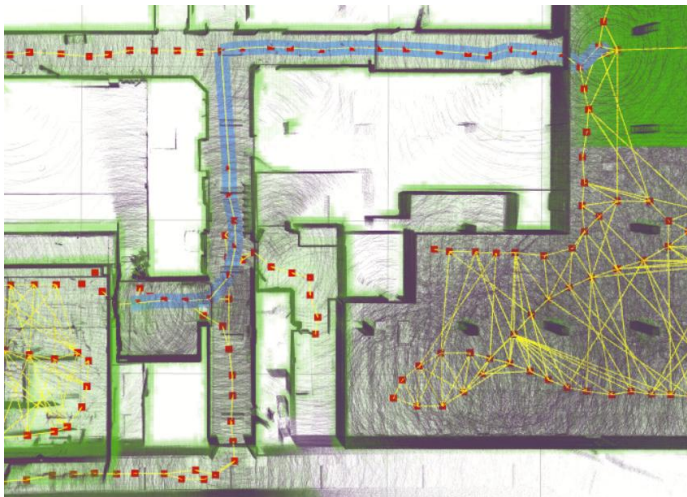
Tracking pedestrians in a
train station



GPS tracks of whales in
Atlantic Ocean

Topological Maps

- Explains connectivity between locations
- May have descriptions associated with places or paths
- Represented as a **graph** of vertices and edges
- Doesn't *necessarily* preserve distances, scale, or directions



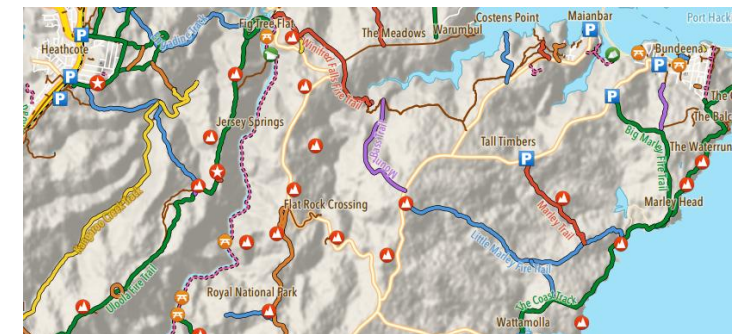
Robot navigation graphs,
e.g. Probabilistic Roadmap (PRM)



Sydney train network



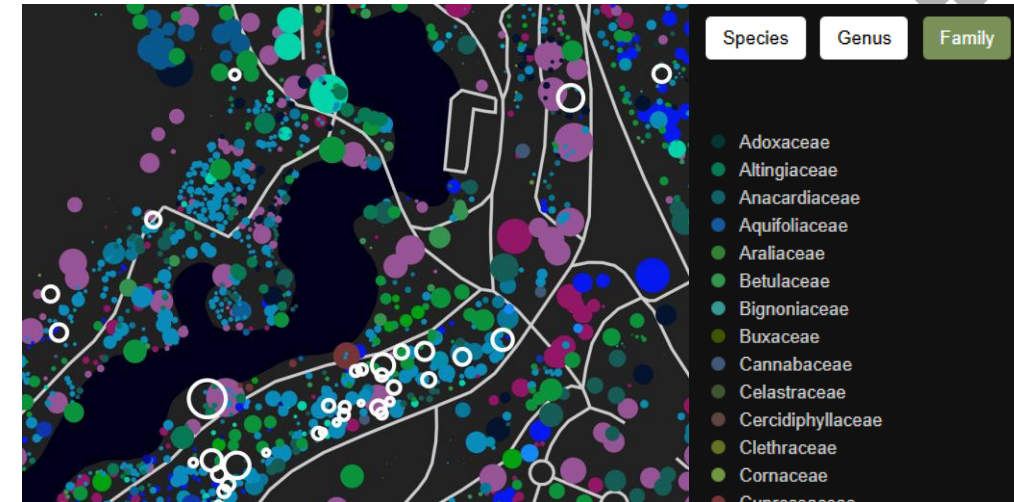
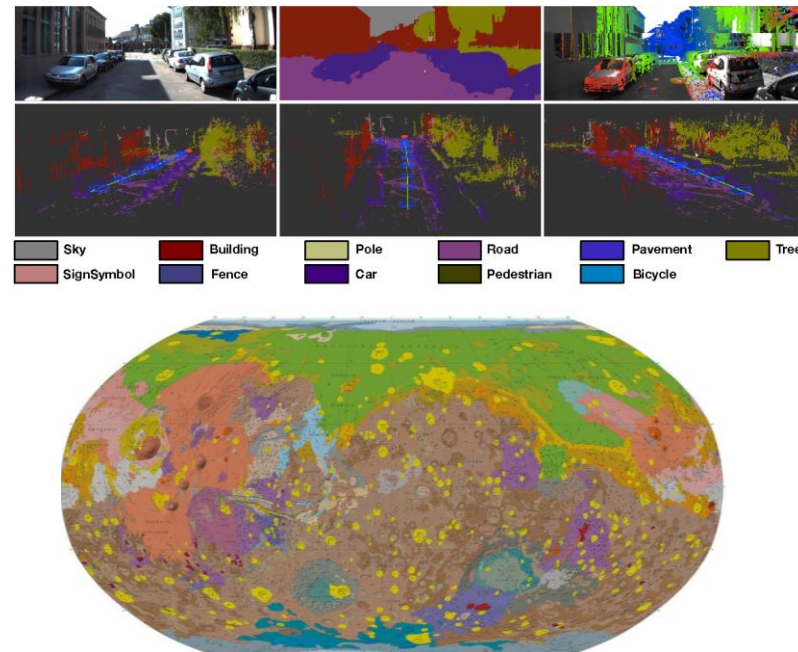
Sydney Botanic Gardens



Royal National Park

Semantic Maps

- Adds descriptions to a map that are “meaningful”
- High level robot commands, e.g. “go to office”
- Human-robot interaction



Map of trees in a park:

jillhubley.com/project/prospectparktrees/



Royal national park

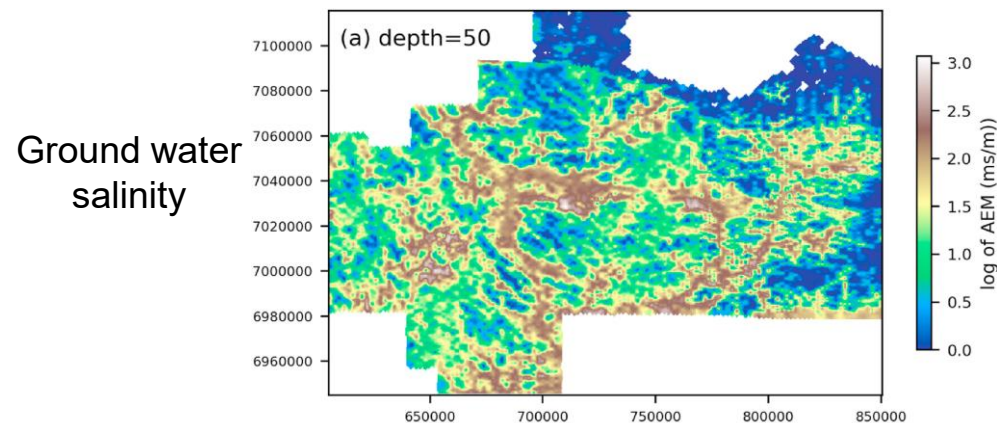
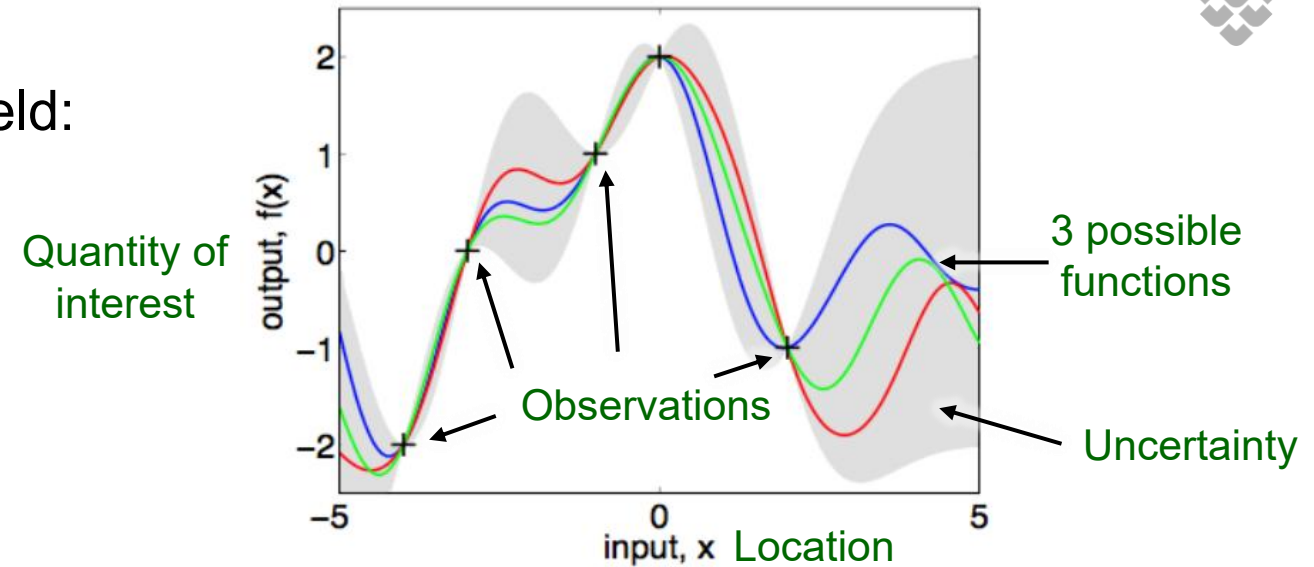
Environmental Phenomena



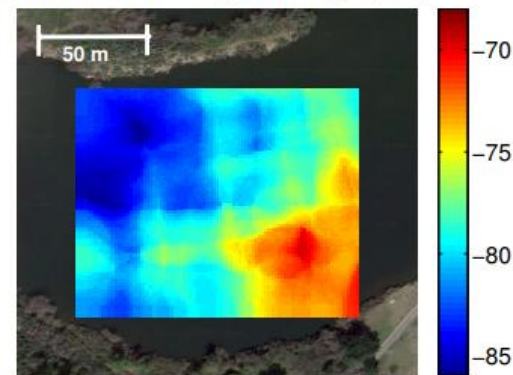
➤ Quantities described as a continuous field:

- Temperature
- Elevation
- Soil type
- Sound volume
- Communication bandwidth

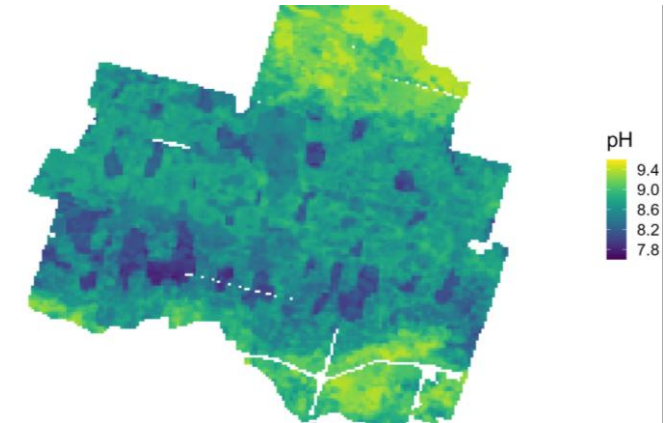
Gaussian Process regression



Predicted Wireless Signal Strength (dBm)

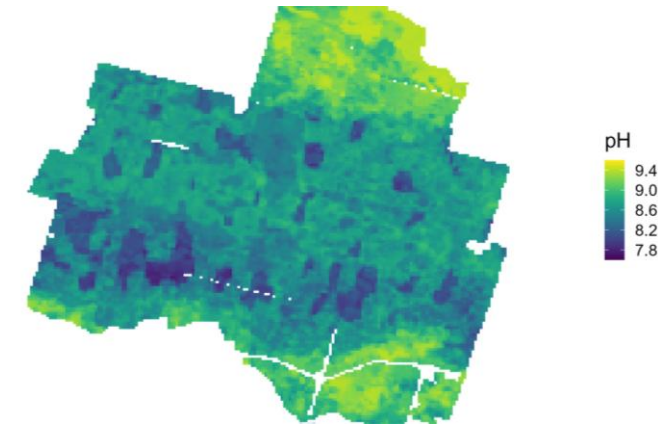
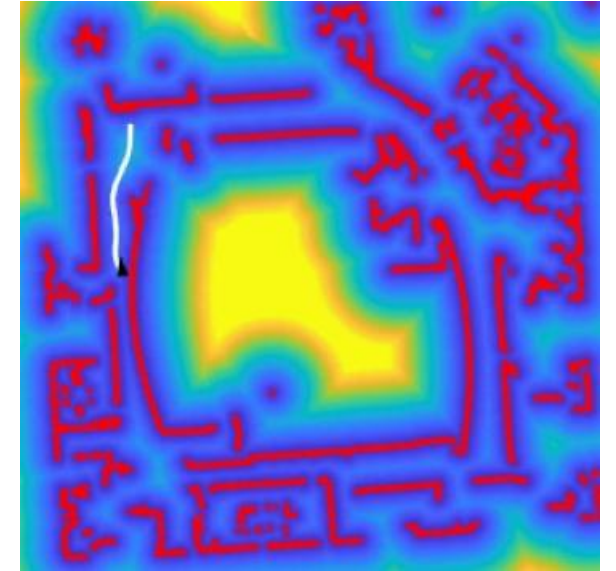
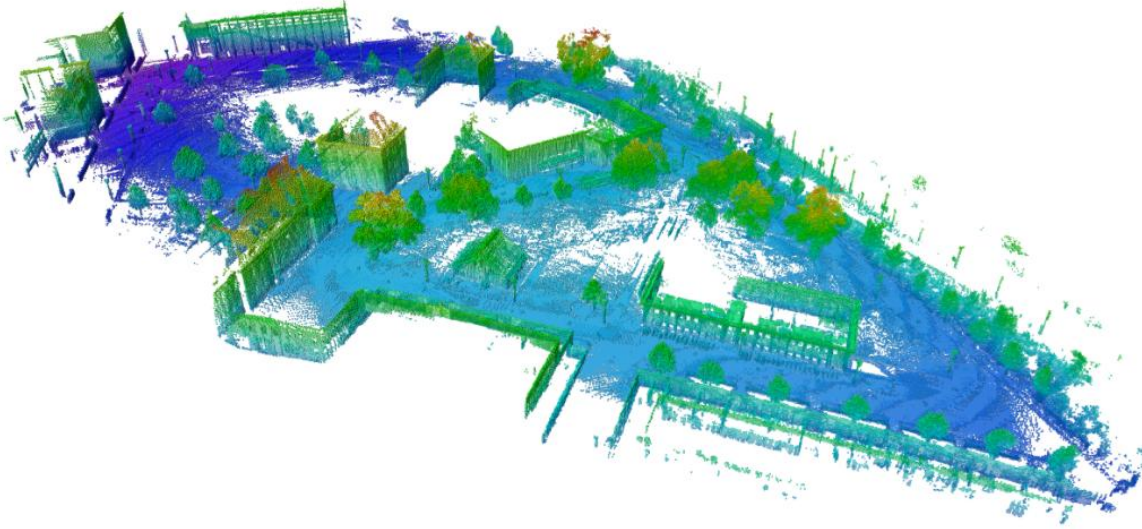


Wireless signal strength



Soil pH level

Which are most meaningful for your project?





Occupancy Grids

Occupancy Grid



- Originally developed for very noisy sensors (sonar)
- Create on-the-fly as a robot moves through an environment
- Need a way to model uncertainty

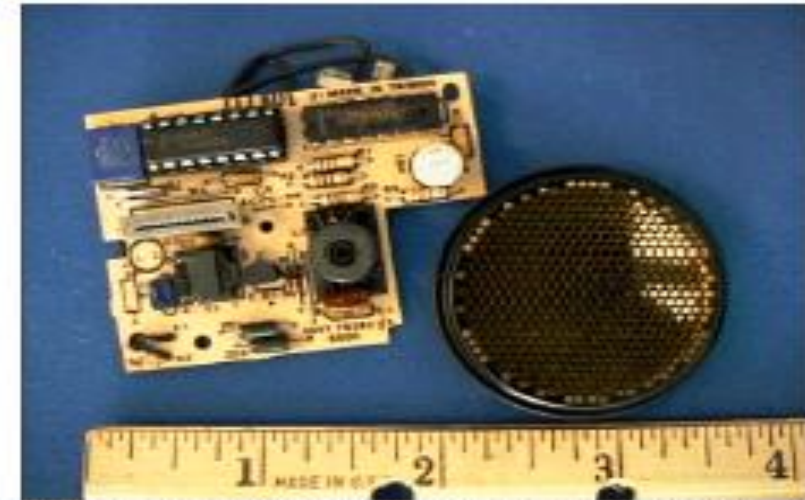
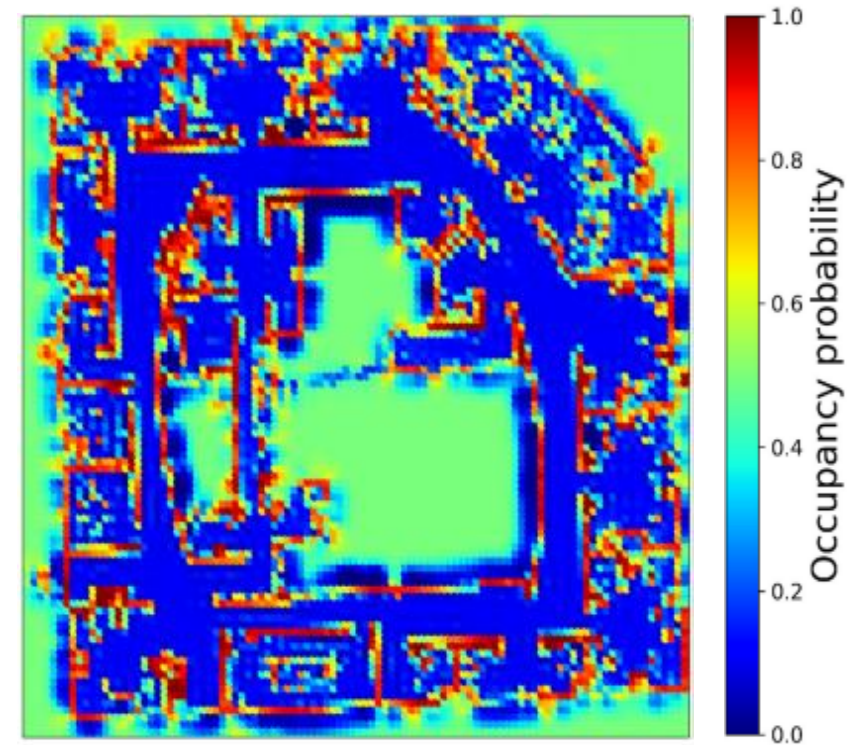
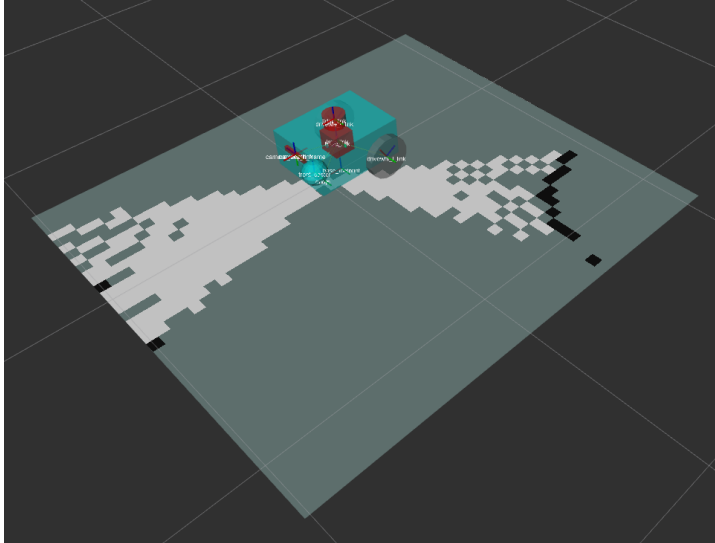


Figure 4.4: The Robot's OBM included the base and a small

Occupancy Probability



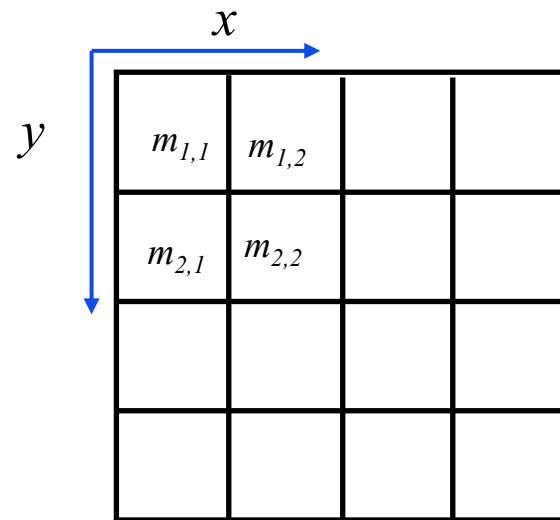
- Each cell is a binary random variable that models occupancy
 - Cell is occupied $p(m) = 1$
 - Cell is not occupied $p(m) = 0$
 - No knowledge $p(m) = 0.5$



Occupancy Grid Map



- Discrete array of cells, each with an occupancy variable
 - $m \rightarrow \{\text{free, occupied}\} \rightarrow \{0,1\}$



Problem: Estimate Map from Sensor Data



Given sensor data $z_{1:t}$ and the poses $x_{1:t}$ of the sensor, estimate the map

$$p(m \mid z_{1:t}, x_{1:t}) = \prod_i p(m_i \mid z_{1:t}, x_{1:t})$$



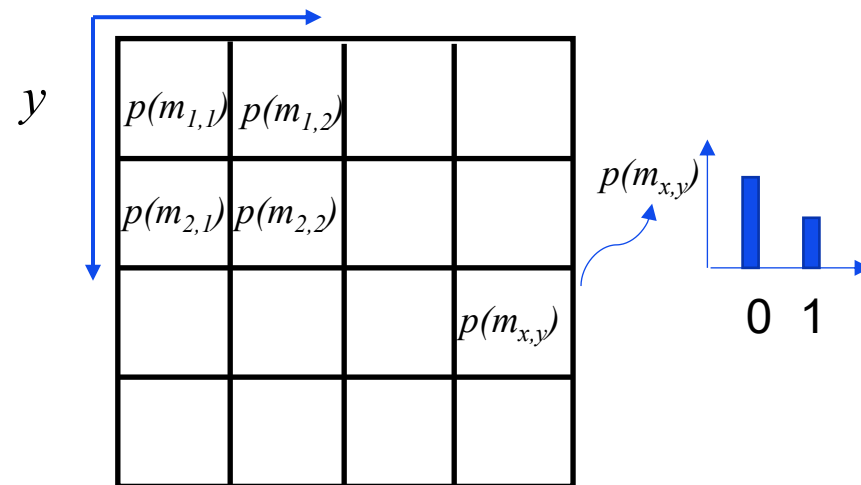
binary random variable

➡ Binary Bayes filter
(for a static state)

Bayesian Filtering



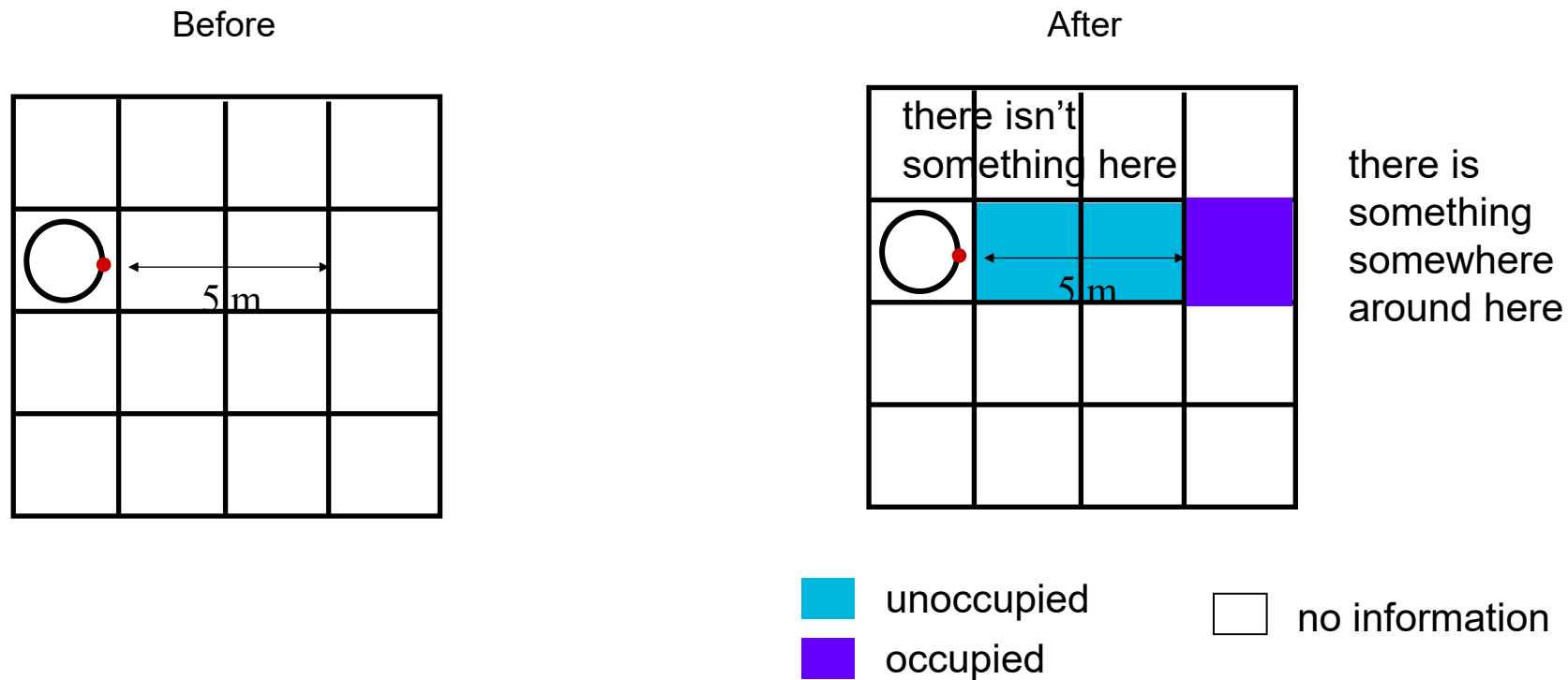
- Recursively update $p(m)$ for each cell



Incorporating an Observation



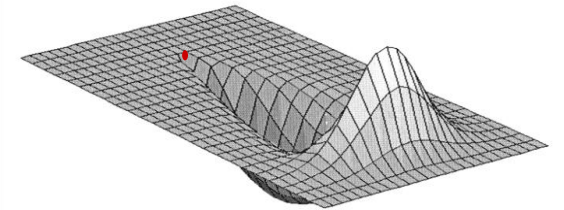
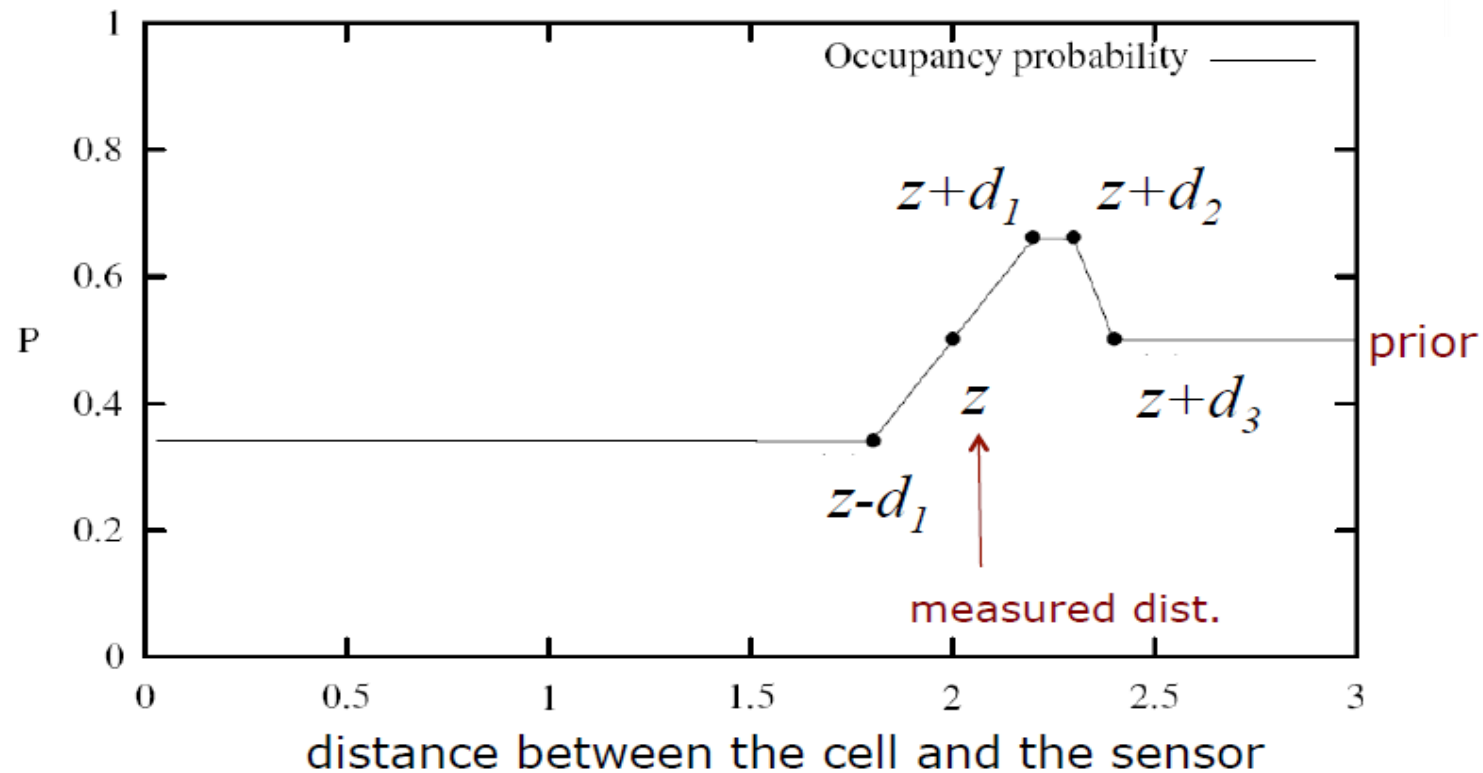
- Let's say our range-finder reports a reading of 5m



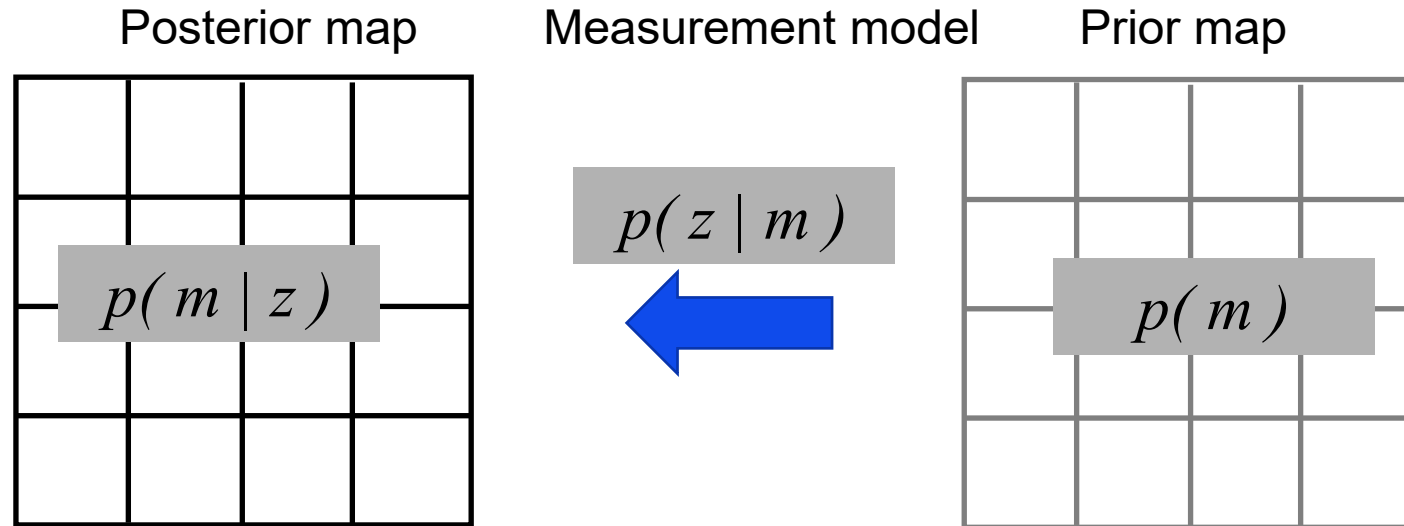
Noisy Observations



➤ Measurement model for a sonar



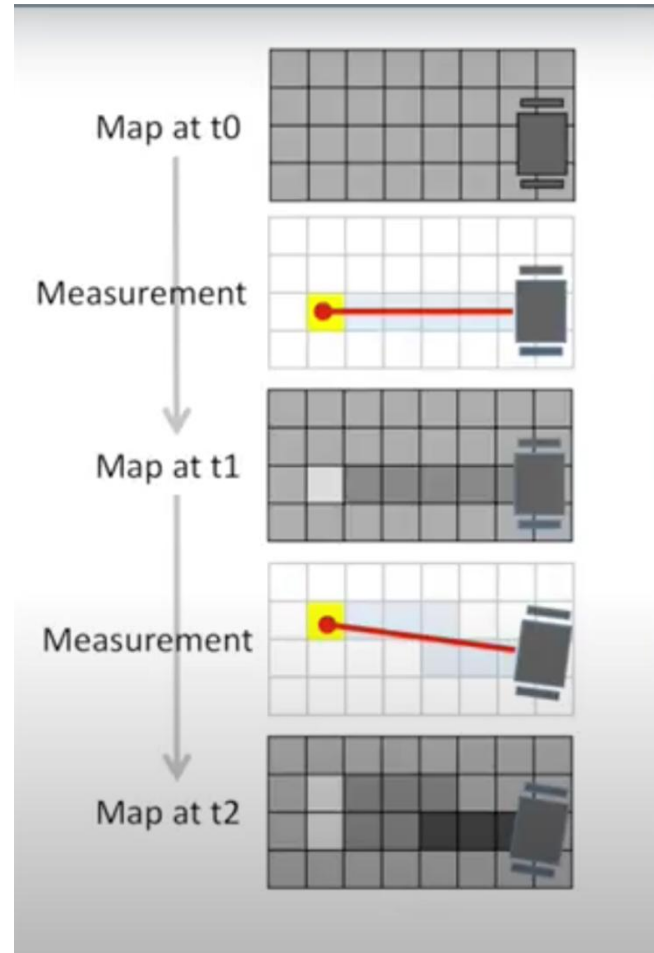
Applying Bayes' Theorem



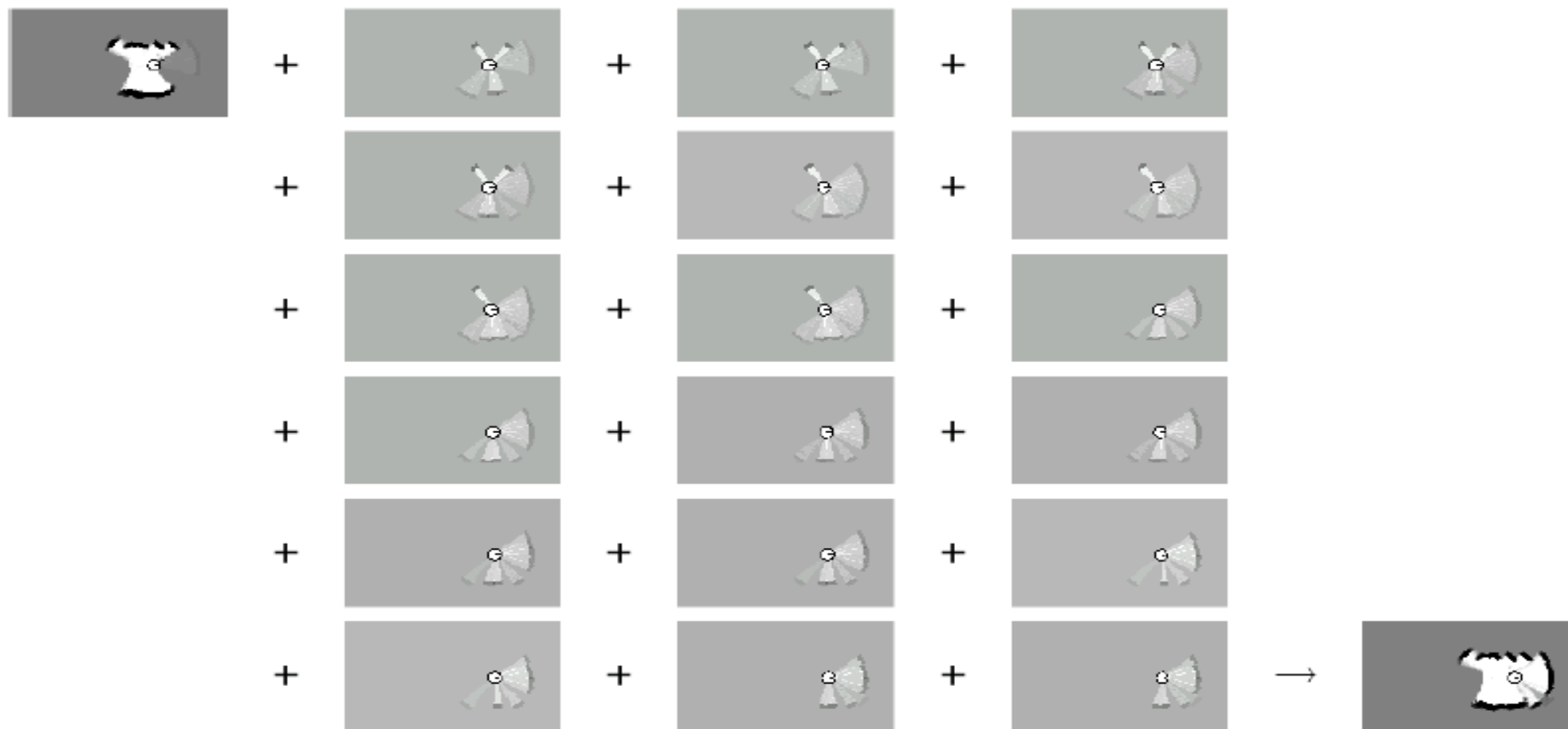
$$p(m | z) = \frac{p(z | m) p(m)}{p(z)}$$

Bayes' rule

Recursive Bayes' Filter



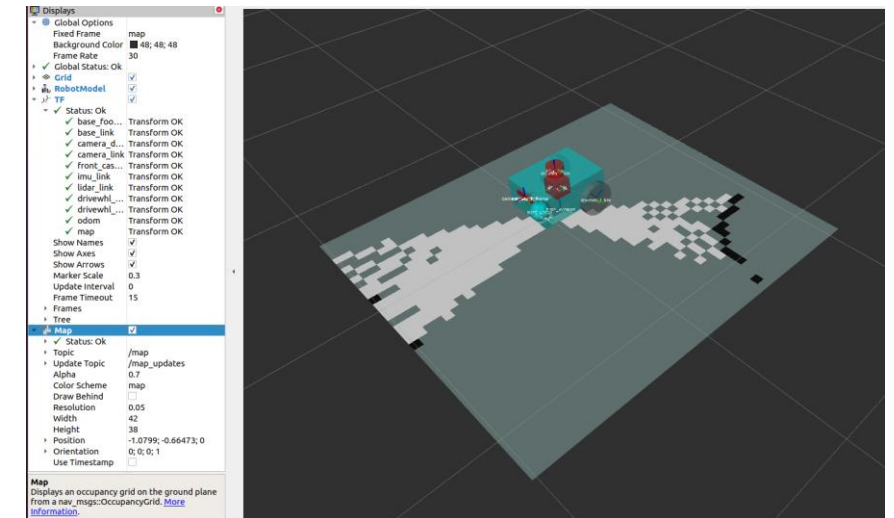
Recursive Bayes' Filter



ROS2 Packages



- Occupancy grids are very common on robots, especially for enabling path planning
- Many common packages, for example see the nav2 package:
https://docs.nav2.org/setup_guides/sensors/mapping_localization.html
- This is used in the 41068 starter package
- This enables basic occupancy grid mapping
- But does **not** provide other types of maps we discussed earlier – for these you will need to look elsewhere, or create them yourself!

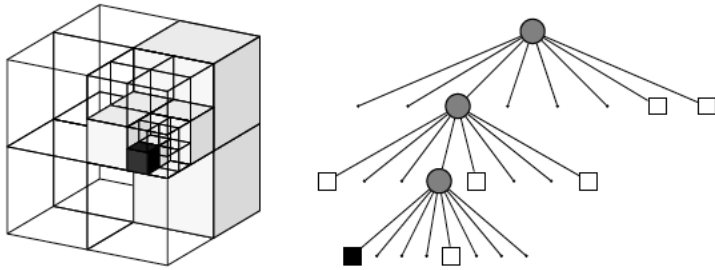


Hierarchical Data Structures

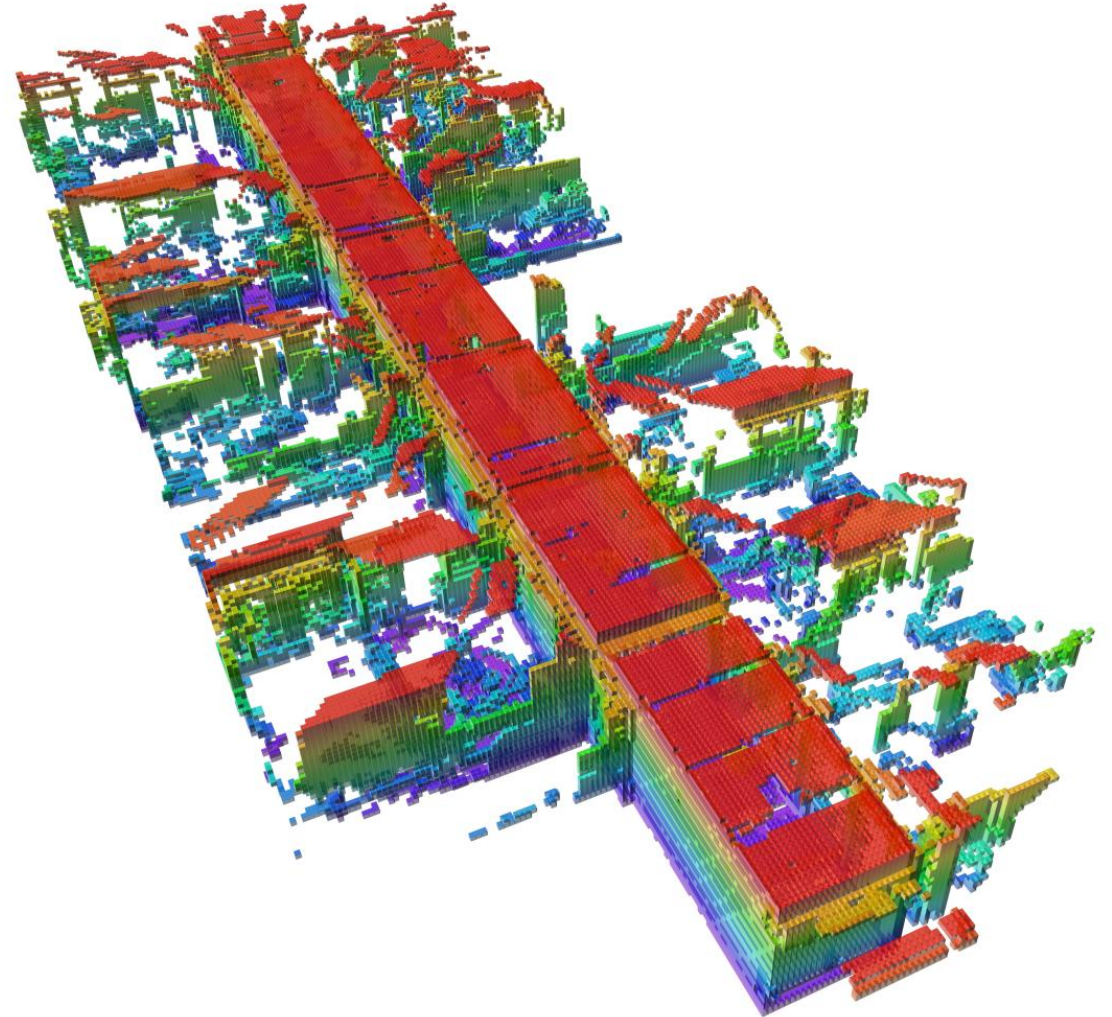


➤ OctoMap 3D mapping library

Tree data structure



Multi-resolution queries



Simultaneous Localisation and Mapping (SLAM)



If we have a map:
We can localise!



NOT THAT SIMPLE!



If we can localise:
We can make a map!



Reflection

Conclusions



- Robots typically have many different sensors
- **Robot mapping** is the process by which a robot builds a representation of its environment using data from its sensors (and prior information).
- Sensors and maps are useful for:
 - **Enabling autonomy**: Understand location of obstacles, landmarks, traversability, navigable areas
 - **Primary tasks**: Provide information to humans and other robots that is specific to the task. May be provided in a **user interface**.
- Many different types of maps for different purposes:
 - Occupancy, terrain, feature, topological, semantic, environmental, ... maps
- Occupancy grid mapping fuses data from multiple noisy observations

Questions for your 41068 Project



- Ideas for different types of maps
- Your project must have some perception/mapping
- Your project must have a user interface

- What types of maps will your robot produce?
 - For primary task
 - For autonomy
- What information will these maps contain?
- How will you show these maps in your User Interface?

